Project Report Format

1. **INTRODUCTION**
    1.1 Project Overview

    The Olympics Games Historical Data Analytics & Interactive Dashboard is a comprehensive data engineering and business intelligence project. It aims to ingest, clean, and structure vast historical Olympic records spanning multiple decades into a relational 7-sheet data framework. By translating complex tabular rows into 2 dynamic interactive dashboards, the project delivers real-time analytical capabilities regarding country-wise medal growth, athlete distributions, and event dynamics.

    1.2 Purpose

    The main purpose is to eliminate data friction for sports analysts, researchers, and fans. Instead of parsing massive, unorganized spreadsheets, users can utilize dynamic filtering paths (Year, Country, Sport) combined with an integrated audio guide narrative to extract strategic historical sports insights within seconds.
2. **IDEATION PHASE**
    2.1 Problem Statement
    2.2 Empathy Map Canvas
    2.3 Brainstorming
3. **REQUIREMENT ANALYSIS**
    3.1 Customer Journey map
    3.2 Solution Requirement
    3.3 Data Flow Diagram
    3.4 Technology Stack
4. **PROJECT DESIGN**
    4.1 Problem Solution Fit
    4.2 Proposed Solution
    4.3 Solution Architecture

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

7. RESULTS

7.1 Output Screenshots

8. ADVANTAGES & DISADVANTAGES

Advantages:

*** Centralized Relational Structure: The 7-sheet architecture prevents data redundancy and speeds up lookup operations.**

*** Dual-Layered Visualizations: Segregates high-level summaries from deep-dive athlete metrics for better clarity.**

*** Low Analytical Barrier: Dynamic dropdown filters allow non-technical users to query the database seamlessly.**

*** High Efficiency: Optimized index paths ensure low dashboard rendering latency and quick query execution times**

Disadvantages:

*** Static Baseline Dependencies: Requires regular manual refreshes if new post-2026 Olympic data points are introduced.**

*** Client-Side Hardware Constraints: High record volumes require standard minimum system RAM (8GB) for glitch-free local parsing**

9. CONCLUSION

The project successfully bridges the gap between raw, scattered sports archives and actionable visual intelligence. By systematically tracking historical data transformations from

ingestion to BI layout presentation, the solution establishes a reliable framework for sports trend visualization. The addition of interactive cross-filters and multi-modal narrative guides ensures high user satisfaction and robust usability metrics.

10. FUTURE SCOPE

*** Live API Integration: Connecting the backend layers directly to official IOC (International Olympic Committee) web streams for automatic real-time data ingestion.**

*** Predictive Analytics Modalities: Introducing machine learning models to forecast upcoming Olympic medal tallies based on historical country training paths.**

*** Advanced Cross-Platform Deployment: Compiling the dashboards into native mobile applications for broader accessibility.**

11. APPENDIX

*** Source Code / Formula Framework: Standard relational spreadsheet schemas, VLOOKUP/INDEX-MATCH scripts, and dashboard query historical logs.**

*** Dataset Link: Kaggle Historical Olympic Games Datasets / Open-source Sports Archives.**